



Effect of the relative differences in the natural frequencies of parallel cable-stayed bridges during interactive vortex-induced vibration

Jin Park^{a,b}, Ho-Kyung Kim^{c,*}

^a Dept. of Safety and Environment Research, The Seoul Institute, 57 Nambusunhwan-ro, 340-gil, Seocho-gu, Seoul 06756, South Korea

^b Korea Bridge Design and Engineering Research Center, Seoul National University, South Korea

^c Dept. of Civil and Environmental Engineering, Seoul National University, Gwanak-ro, Gwanak-gu, Seoul 08826, South Korea

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ABSTRACT

This study was focused on the effects of vortex-induced vibration (VIV) in parallel twin cable-stayed bridges. A series of wind tunnel tests were performed to investigate the effect of the relative differences in the setup of the natural frequencies of parallel decks during VIV. The differences in the natural frequencies of the two decks are represented in terms of the frequency ratio (FR) as defined by the ratio of the natural frequency of the upstream deck to that of the downstream deck. Six FRs were examined to cover all the potential ranges of relative differences in the natural frequencies of the two decks. The interactive motions of the two decks were examined with normalization parameters to demonstrate the fundamental mechanics of complicated VIV of parallel twin cable-stayed bridges from the viewpoints of fluid and structural dynamics.

1. Introduction

Owing to a series of successful planning and construction projects for long-span bridges around the world over the past three decades, knowledge on bridge aerodynamics has broadened considerably. The shapes of bridge decks have evolved for the suppression of flutter instability even for bridges with main spans of more than 3 km. Even though one could carefully state that the modern prototypes of aerodynamic deck sections are prepared for flutter-related issues, the issue of vortex-induced vibration (VIV) is case dependent, and continues to command attention.

Recently, VIV in parallel bridges has drawn attention because of the complicated interactions between the decks. The potential for VIV in parallel bridges was described by Kimura et al. (2008), but in-depth investigations followed only after the actual observation of VIV in the upstream deck of parallel cable-stayed bridges in 2011. Seo et al. (2013) successfully reproduced VIV in a wind tunnel and demonstrated that VIV was magnified by the parallel arrangement of decks. In a follow-up study, Kim et al. (2013) successfully demonstrated the existence of interactive VIV based on the field monitoring of data obtained from the bridges and described the low-level inherent damping ratio in the upstream deck as another potential contribution to high-level vibrations. Meng et al.

(2011), Argentini et al. (2015), and Dallaire et al. (2016) have also reported studies of the interactive VIV of parallel decks.

Since one of the parameters in this sort of vibration is the distance of the gap between the two decks, Park et al. (2017) demonstrated that gap distances of five to seven times the depth of the upstream deck critically affected the interactive VIV in parallel cable-stayed bridges, which agreed with the results found by Kimura et al. (2008). Park et al. (2017), however, successfully demonstrated how the vulnerable gap distance was closely related to the moving distance of one vortex during a period of deck oscillation.

Another potential parameter that can play an important role in interactive VIV would be the relative differences in the natural frequencies of the two decks. For the VIV in the 2011 study, the natural frequency of the upstream deck was a bit lower than that of the downstream deck. According to Kim et al. (2013), the complicated interactions that can occur between two decks have been identified in operational field monitoring as well as in a wind tunnel and seem to be critically influenced by the relative differences in the frequencies of both decks.

Based on these results, we examined the interactive VIV in parallel decks by focusing on the relative differences in the natural frequencies of each. The examined frequency difference covered all the meaningful cases so that the fundamental mechanism of the interactive VIV could

* Corresponding author.

E-mail addresses: jinpark@si.re.kr (J. Park), hokyungk@snu.ac.kr (H.-K. Kim).

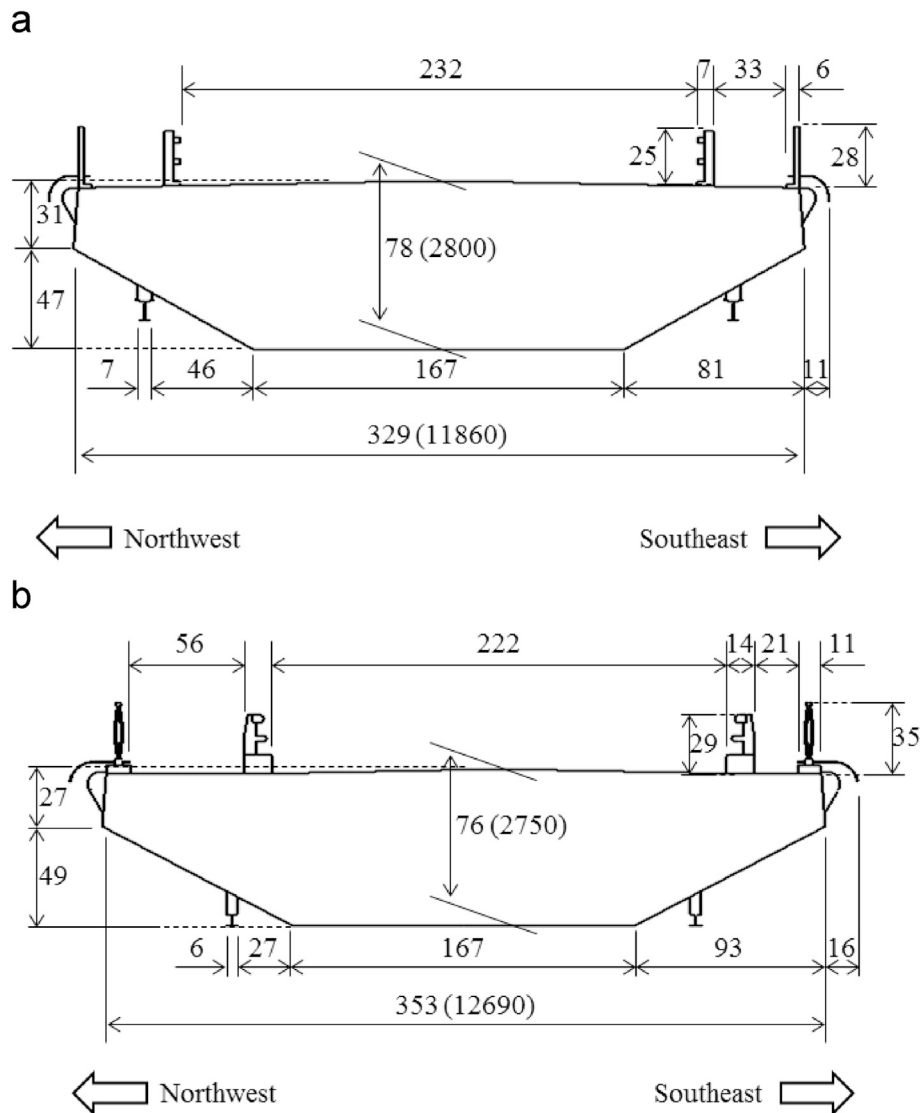


Fig. 1. Dimension (mm) of section models (Values in parentheses represent the dimensions for the prototype): (a) B1 in downstream and (b) B2 in upstream.

be identified.

2. Experimental set-up

The investigated bridges had parallel cable-stayed decks exposed to crosswinds. The downstream and upstream decks will be referred to as B1 and B2, respectively, because B1 was built first in 1984 and B2 was later added in 2005, which fulfilled a concept for twin bridges. A northwestern wind prevails in the area of these bridges.

The two sectional models for the wind tunnel tests were identical to those used in several of the previous studies (Seo et al., 2013; Kim et al., 2013; Park et al., 2017). The sectional models were fabricated of balsa with the following length scale: $\lambda_L = L_m/L_p = 1/36$, where L_m and L_p represent the lengths of the model and prototype, respectively. The

length of the model was 0.9 m, which was equivalent to a length of 32 m for the prototype. Both decks were equipped with barriers, handrails, inspection rails, guide vanes, and cable anchorage. All details were correctly reflected in the model to a consistent scale. Each sectional model was equipped with two end plates 0.40 m in width and 0.13 m in height for securing two-dimensional flow conditions along the sectional model. The geometries and dimensions of the sectional models are shown in Fig. 1.

The two decks were placed in a wind tunnel, as shown in Fig. 2. B2 was set in the upstream position. The center-to-center distance between the decks was 618 mm, which corresponded to an as-built distance of 22.25 m for the prototype. This setup conformed to the conditions of the unexpected VIVs in the 2011 study, which prompted an investigation of the bridges (Seo et al., 2013). The two sectional models were each

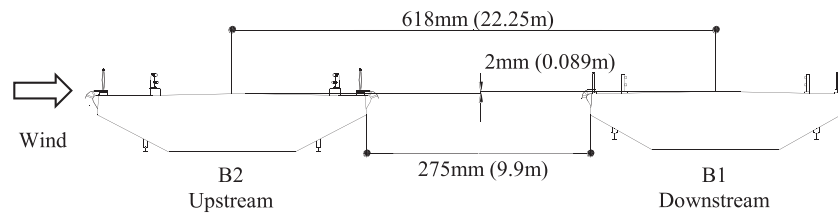


Fig. 2. Disposition of the two decks in the wind tunnel (Values in parentheses represent the dimensions of the prototype).

Table 1
Similitude law and setup parameters for sectional models.

Parameters	Similarity scale	B2		B1	
		Prototype	Model	Prototype	Model
Width (B) [m]	$\lambda_L = 1/36$	12.690	0.353	11.860	0.329
Depth (D) [m]	λ_L	2.750	0.076	2.800	0.078
Mass per unit length (m) [kg/m]	λ_L^2	8978	6.988	6950	5.359
Mass moment of inertia per unit length (I_m) [$\text{kg} \times \text{m}^2/\text{m}$]	λ_L^4	152836	0.090	84030	0.049
Vertical damping ratio (ξ) [%]	1	0.2–0.3	0.1	0.5–0.7	0.1
Scruton number for vertical motion ($= 2\pi\xi m/\rho B^2$, $\rho = 1.25 \text{ kg/m}^3$)	1	0.56–0.84	0.28	1.24–1.74	0.25

Table 2
Frequency ratios and setup frequencies in the wind tunnel.

$FR = f_2/f_1$	f_2 (Hz)	f_1 (Hz)
0.63	4.01	6.38
0.86	4.01	4.64
0.94	5.26	5.56
1.00	5.26	5.26
1.08	4.01	3.72
1.96	4.01	2.04

mounted on spring-supported systems that permitted both vertical and torsional motions, although only vertical VIVs were taken into consideration in this study. The setup parameters are summarized in Table 1.

The natural frequency of B2 was 0.436 Hz on the prototype scale, and that of B1 was slightly higher at 0.513 Hz. Accordingly, the frequency ratio for the two decks was 0.436 Hz/0.513 Hz = 0.85. Since this study examined the effect of the frequency ratio of the interactive VIV of two parallel bridges, both the definition and consideration of the range-of-frequency ratio are important. The differences in the natural frequencies of the two decks are represented in terms of the frequency ratio (FR) as defined by Eq. (1).

$$FR = \frac{\text{frequency of upstream deck}}{\text{frequency of downstream deck}} = \frac{f_2}{f_1} \quad (1)$$

In the present study, six FRs were considered: 0.63, 0.86, 0.94, 1.00, 1.08, and 1.96. A FR = 0.86 conformed to the as-built condition of the investigated bridges. The setup frequencies of the sectional models in the actual wind tunnel tests are summarized in Table 2. As shown in Table 2, the frequency of B2 was basically fixed at 4 Hz and that of B1 was changed to achieve the target FR. However, a slightly different frequency of 5.26 Hz was used in B2 for FR = 0.94 and 1.00. The damping ratios of both decks were maintained to a minimal level of 0.1% for a better observation of the vibration.

At this point, it is worth mentioning that the current examination of interactive VIV for different FRs covers the effect of different wind directions when the similar shapes of two decks are taken into consideration.

3. Interactive motions between the two decks for examined FRs

3.1. Variation of A-V curves for different FRs

Fig. 3 shows the changes in amplitude-velocity (A-V) curves of the two decks for different FRs. The vertical motions were monitored in terms of root mean square (RMS) values when the vortex-induced vibration was fully developed in the dominant deck. Since the lock-in wind velocities for both decks varied depending on the actual setup frequencies of the decks, as shown in Table 2, most of the results were represented in terms of reduced wind velocity for the sake of data analysis conformity. For example, the abscissa in Fig. 3 was normalized with the depth (D_2) and natural frequency (f_2) of B2 for both decks. The RMS responses of vertical motions were also normalized in terms of D_2 for both decks, as shown in Fig. 3.

According to Fig. 3, the range of lock-in wind velocity of B2 was fixed around 9–12 regardless of FR, while that of B1 was changing. The fixed range of reduced wind velocity in B2 was simply due to the reduced wind velocity being normalized in terms of the natural frequency of B2 (f_2), but it was noteworthy that the motions of B2 were not much influenced by FR in terms of either amplitude or range of lock-in wind velocity. However, the amplitudes of B1 were greatly affected by FR, as shown in Fig. 3.

3.2. Cases of FR = 0.63 and 1.96

In the cases of FR = 0.63 and 1.96, the lock-in wind velocities of the two decks were completely separate, as shown in Fig. 3(a) and (f). Even though the sequence of VIV in the two decks was reversed, the amplitudes of the VIV in both decks were basically identical. This implies that when the ranges of lock-in wind velocity for two decks are sufficiently separated by a significant difference in natural frequencies, each deck will show an independent motion.

3.3. Cases of FR = 0.86 and 0.94

Increasing the FR from 0.63 to 0.94 approximated close natural frequencies in both decks and the lock-in wind velocities for the two decks were overlapped. Accordingly, not only the range of lock-in wind velocity but also the amplitude of B1 was critically influenced. In the case of

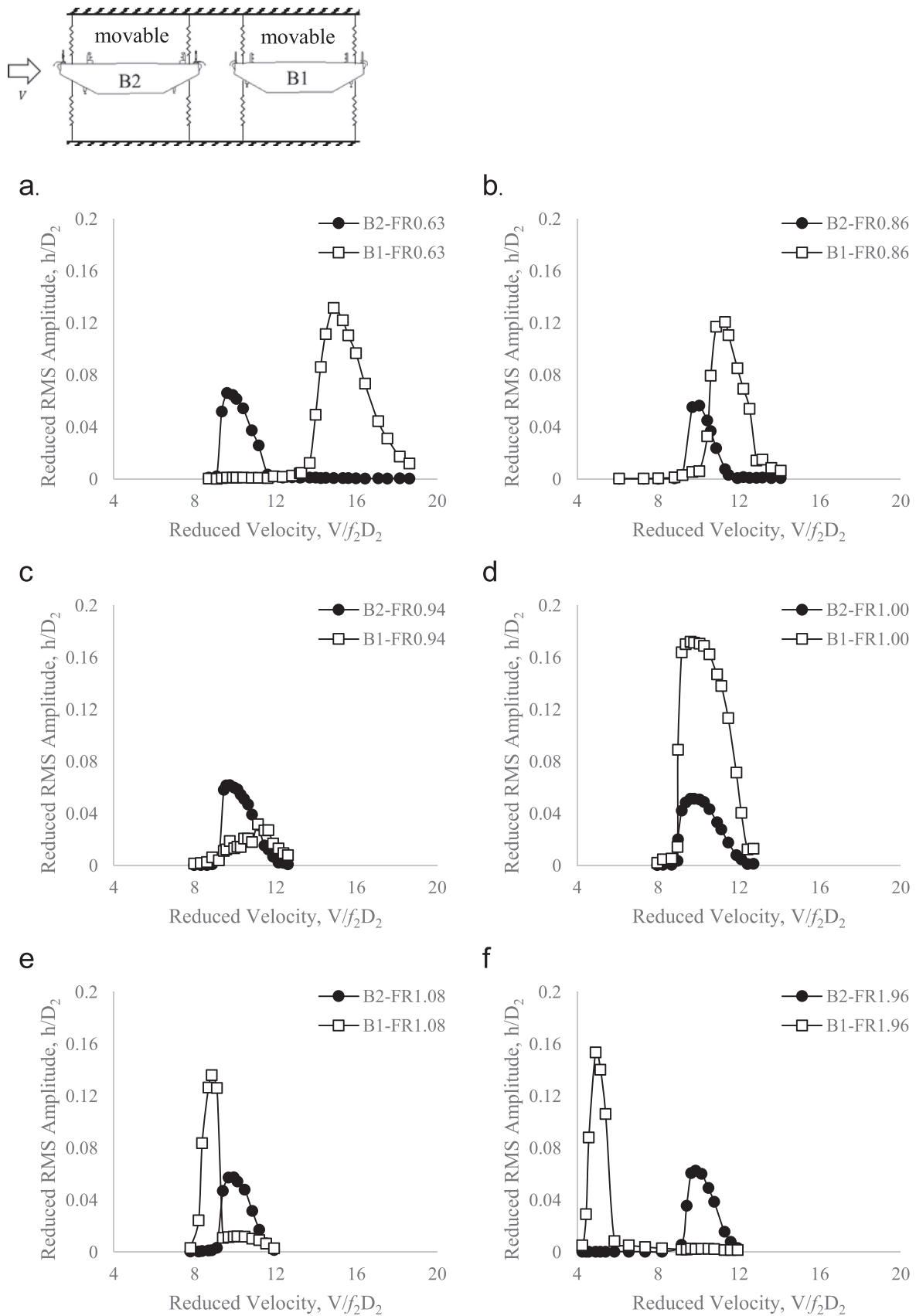


Fig. 3. Bridge responses according to the wind speeds for (a) FR = 0.63, (b) FR = 0.86, (c) FR = 0.94, (d) FR = 1.00, (e) FR = 1.08, and (f) FR = 1.96.

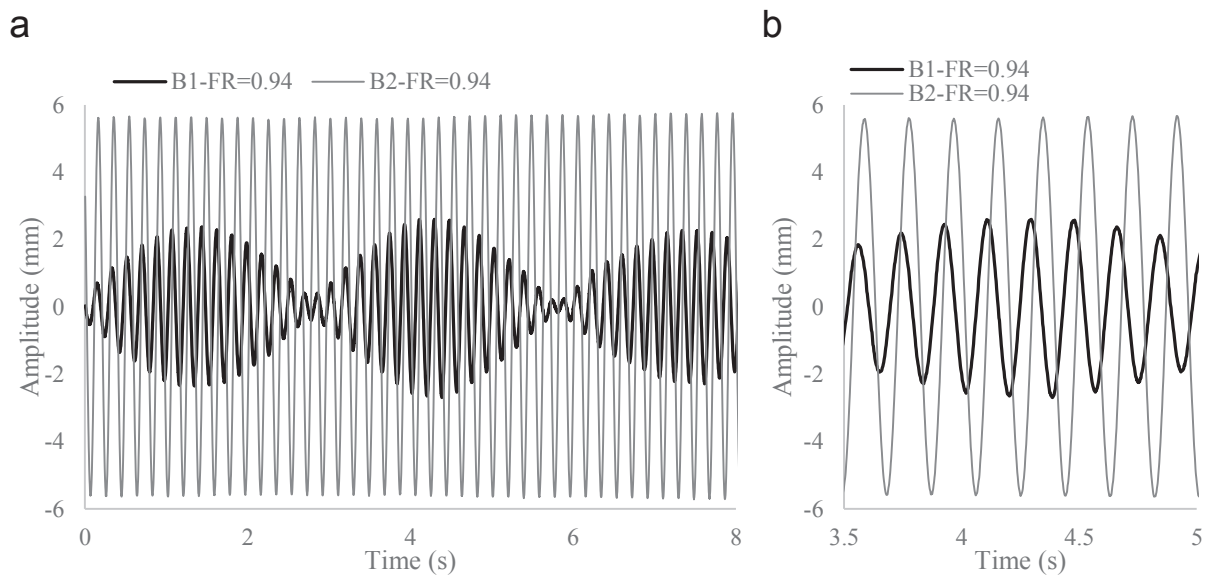


Fig. 4. (a) Time history of the vertical displacements and (b) zoom image at $V/f_2D_2 = 9.6$ and $FR = 0.94$.

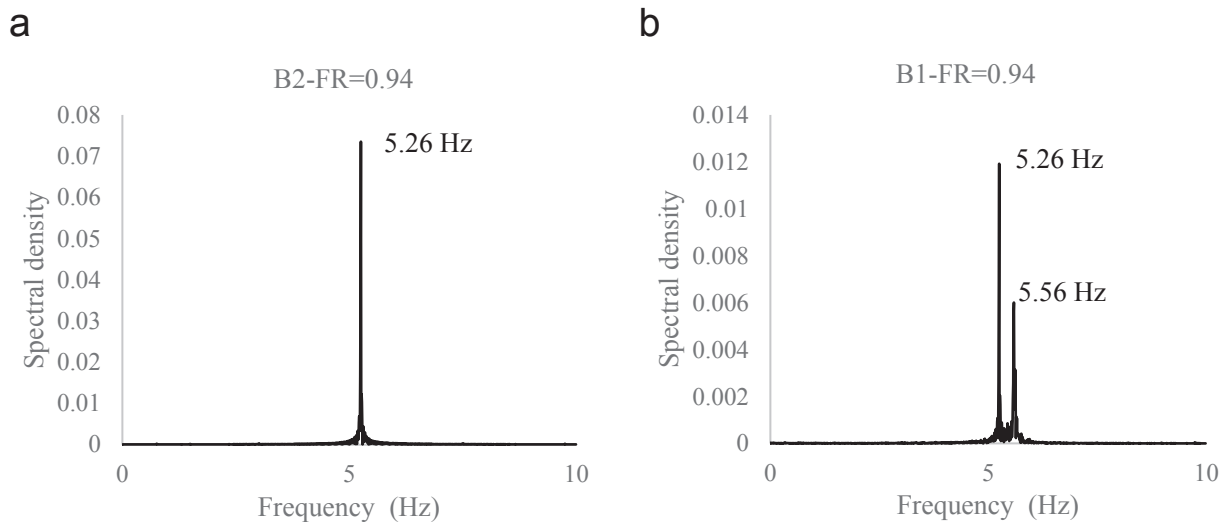


Fig. 5. Spectral densities of the normalized vertical displacements of (a) B2 and (b) B1 at $V/f_2D_2 = 9.6$ and $FR = 0.94$.

$FR = 0.86$, the VIV started first in B2 and was transferred to B1 as the wind velocity increased. While B2 reached the maximum VIV, the vibration in B1 was suspended even for the potential lock-in conditions in terms of wind velocity. When $FR = 0.86$, this in fact approximated the as-built condition of the bridge, which was thoroughly investigated via wind tunnel testing and field monitoring in previous research (Seo et al., 2013; Kim et al., 2013; Park et al., 2017).

As the FR reached 0.94, the lock-in velocities of the two decks were almost identical, and the motion of B1 was critically disturbed by the motion of B2, which resulted in the minimum amplitude among the observed cases. Figs. 4 and 5 show the time histories of the motions and

the motional frequency components of both decks at a reduced wind velocity of 9.6. B2 showed a larger amplitude than B1 at this wind velocity. Beat phenomena were observed for B1 due to two adjacent frequency components in motion - one was tuned to the natural frequency of B2, which demonstrated the motional frequency component of B1 excited by the lock-in vortex trails generated from B2, while the other was tuned to the natural frequency of B1 itself. This beat motion was well reported by Kim et al. (2013) for the as-built condition of a bridge with a $FR = 0.86$. For an increased wind velocity of 11.1 in reduced form, the time histories of the motions and the motional frequency components of both decks are shown in Figs. 6 and 7. The amplitude of B1 increased and

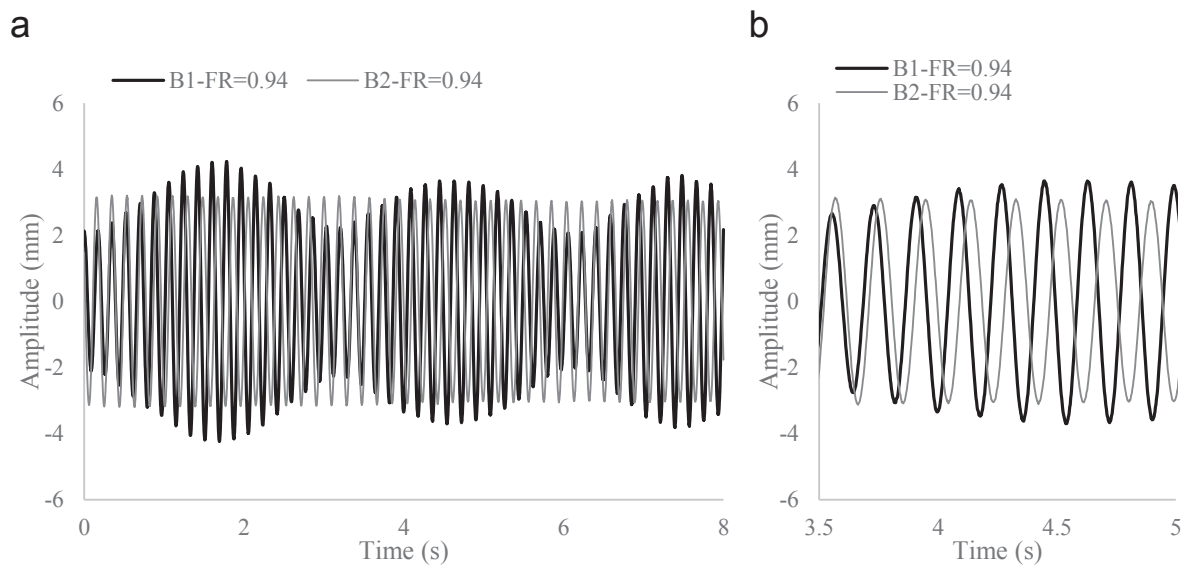


Fig. 6. (a) Time history of the vertical displacements and (b) zoom image at $V/f_2D_2 = 11.1$ and $FR = 0.94$.

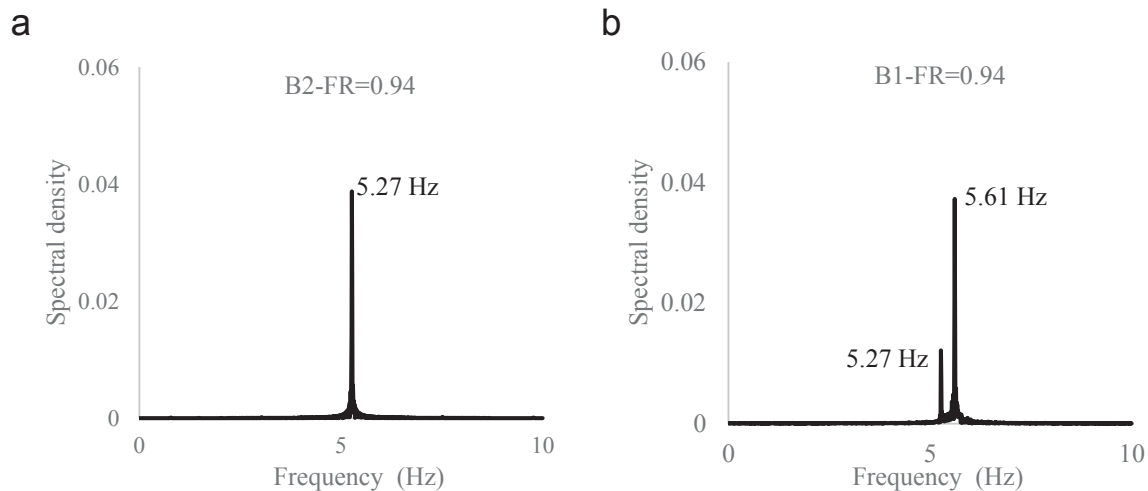


Fig. 7. Spectral densities of the normalized vertical displacements of (a) B2 and (b) B1 at $V/f_2D_2 = 11.1$ and $FR = 0.94$.

exceeded that of B2. Two components of motional frequencies were still observed in B1, which again resulted in a beat motion.

3.4. Case of $FR = 1.0$

$FR = 1.0$, as shown in Fig. 3(d), was a unique case. The VIV in B1 reached the maximum amplitude from among the observed cases. The time histories of the motions in both decks are shown in Figs. 8 and 9. Since the two decks were vibrating with the same natural frequencies, as shown in Fig. 10, the motion of B1 was eventually tuned to the motion of B2 in a phase whereby each experienced the ultimate interactive vibration in terms of amplitude.

3.5. Case of $FR = 1.08$

When FR was increased above a value of one, vibration was observed first in B1, as shown in Fig. 3(e). Figs. 11 and 12 show the time histories of the motions and the motional frequency components of both decks at a reduced wind velocity of 9.1. Each deck showed motion tuned to its own natural frequency, as shown in Fig. 12.

As the wind velocity increased further, B2 entered into the lock-in phenomenon and was subjected to VIV at a reduced wind velocity of 9.4. At this point, the VIV in B1 was suddenly decreased due to the motion-induced vortices of B2, as shown in Fig. 3(e). Figs. 13 and 14 show the time histories of the motions and the motional frequency components of both decks at a reduced wind velocity of 9.4. The

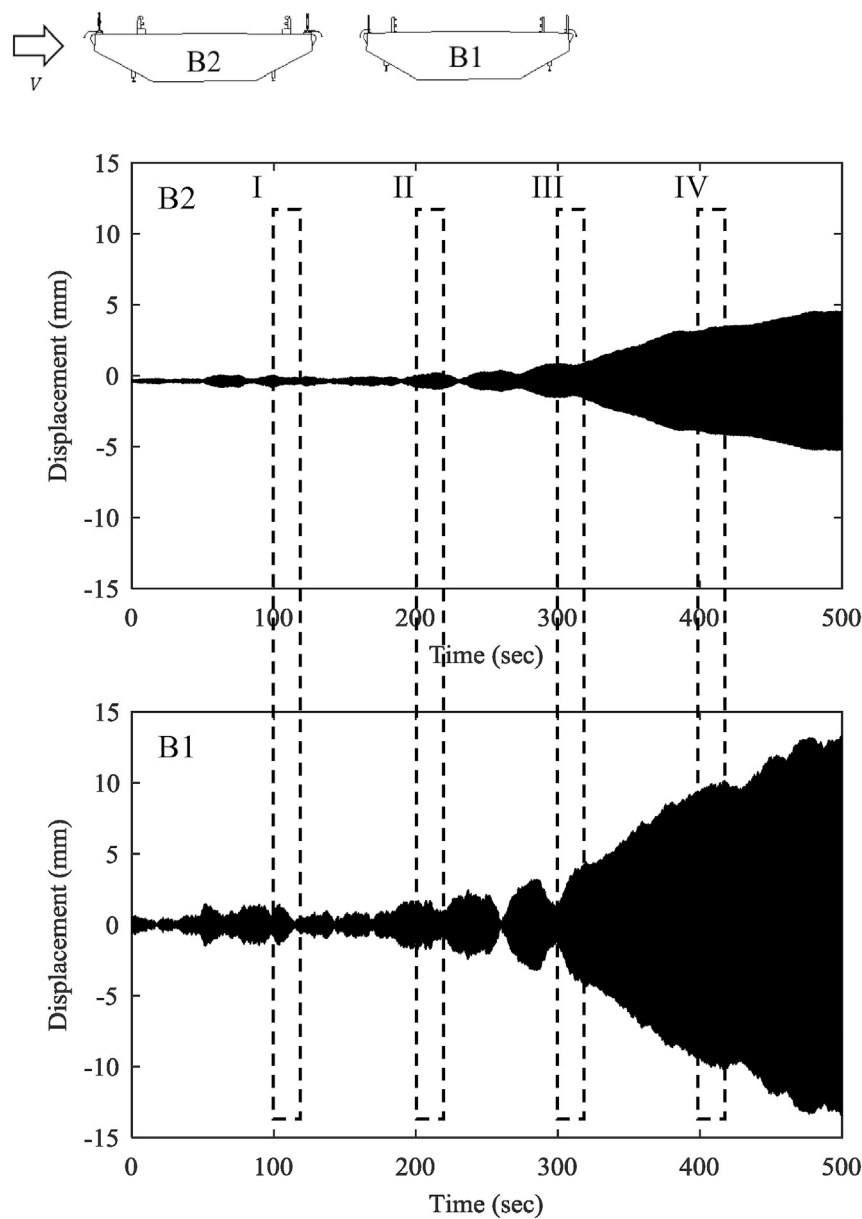


Fig. 8. Time histories for $FR = 1.00$.

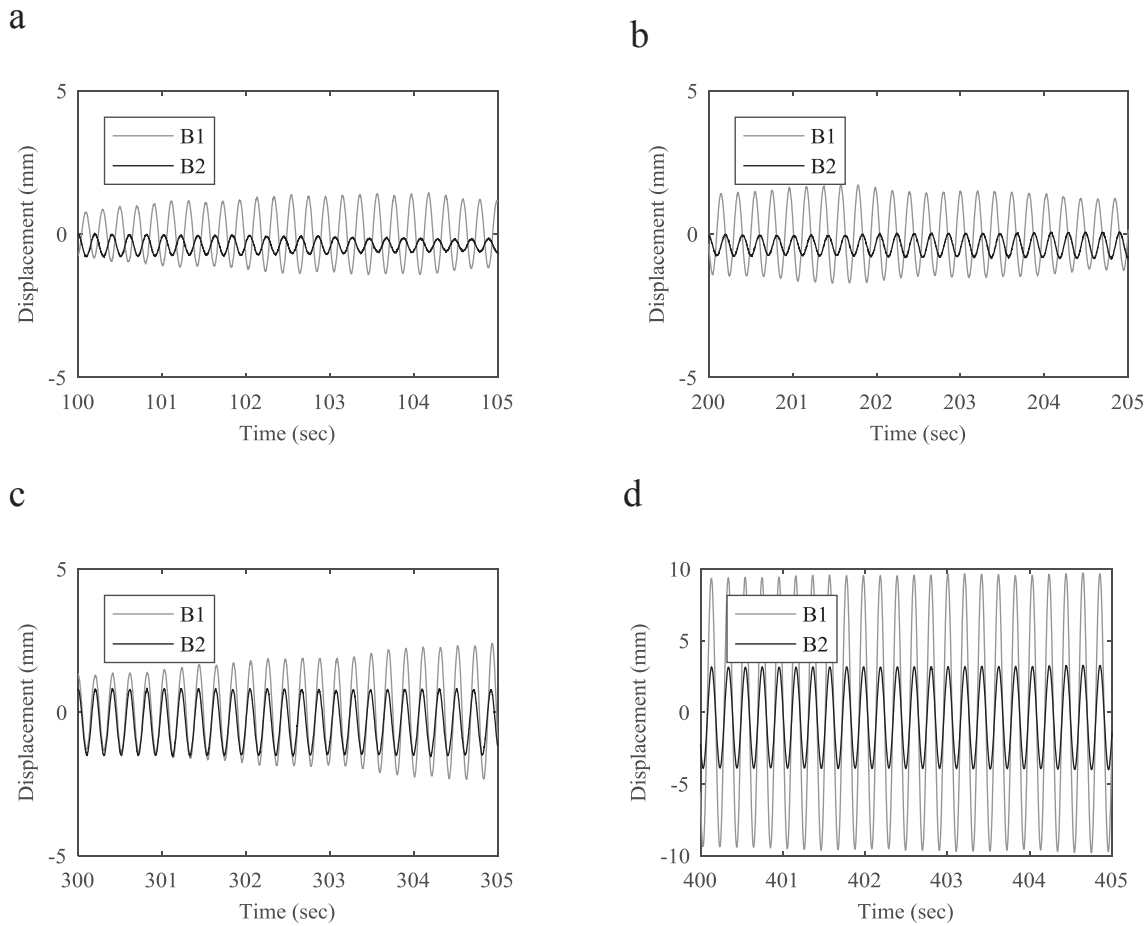


Fig. 9. Time histories for (a) region I, (b) region II, (c) region III, and (d) region IV in Fig. 4.

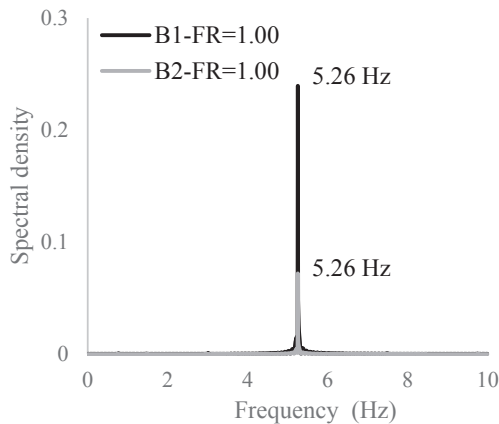


Fig. 10. Spectral densities of the normalized vertical displacements at $V/f_2 D_2 = 9.6$ and $FR = 1.00$.

closeness of these two frequency components again led to the beat phenomena in B1.

3.6. Summary of the interactive motions for different FRs

Fig. 15 shows the onset wind velocities of VIVs for both decks. The reduced wind velocities in Fig. 15 were normalized in terms of the natural frequency of each deck. As shown in Fig. 15 (a), the onset wind velocities of B2 were not influenced by the change in FR. The distribution of the onset wind velocities of B1 was also uniform with the exception of $FR = 0.94$, which showed a slightly increased value.

Fig. 16 also shows the A-V curves, normalized with the natural frequency of each deck, for the examined FRs. The effect of FR on the VIVs of each deck was clarified via normalization with the reduced velocity of the natural frequency of each deck. The normalized motions of the VIVs in B2 almost coincided regardless of FR, as shown in Fig. 16 (a), while those for B1 varied in amplitude and in the range of lock-in wind velocity, as shown in Fig. 16 (b).

However, it is worth mentioning that even though B2 was not much influenced by the motion of B1, the existence of B1 was critical in causing the magnified VIV in B2. For reference, the VIV of B2 alone is shown with a dashed line in Fig. 16 (a). This explanation is also valid for B1, as shown in Fig. 16 (b).

Since the motion of B1 was significantly influenced by that of B2, additional wind tunnel tests were performed restraining the motion of B2. As shown in Fig. 17, the motion of B1 then showed a uniform pattern regardless of FR. This implies that the complicated interactive VIV in B1 was induced by the motion-induced vortices generated by B2.

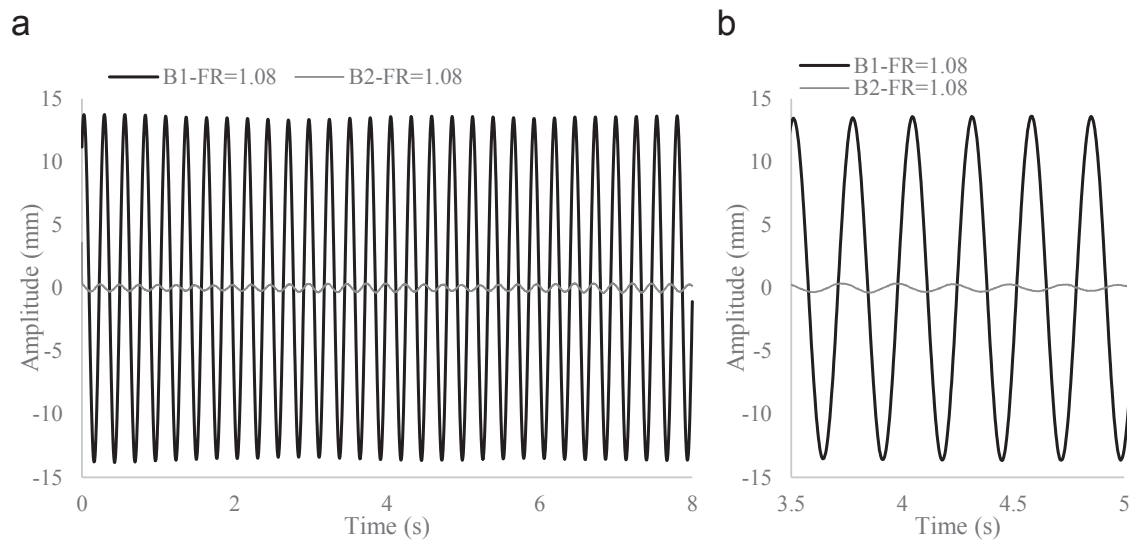


Fig. 11. (a) Time history of the vertical displacements and (b) zoom image at $V/f_2D_2 = 9.1$ and $FR = 1.08$.

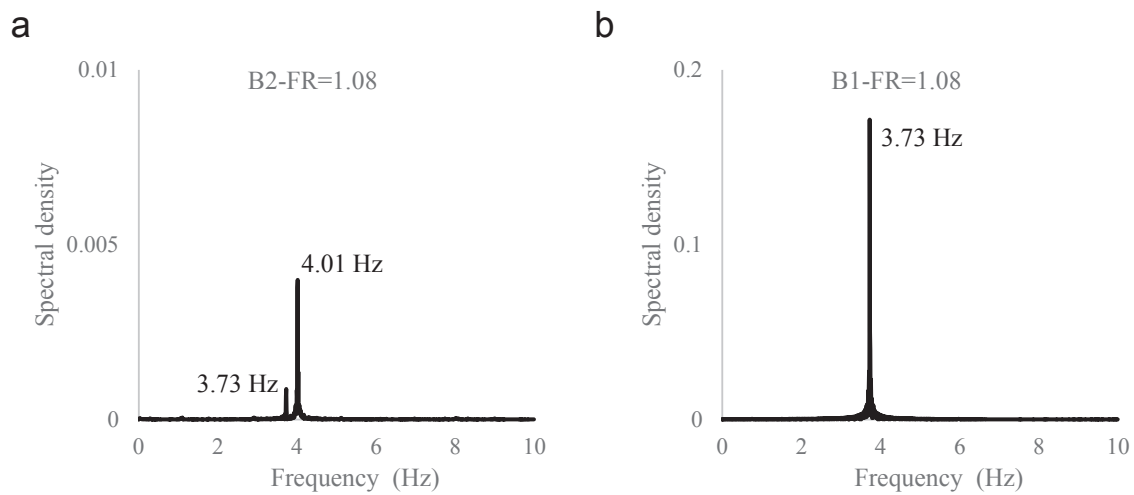


Fig. 12. Spectral densities of the normalized vertical displacements of (a) B2 and (b) B1 at $V/f_2D_2 = 9.1$ and $FR = 1.08$.

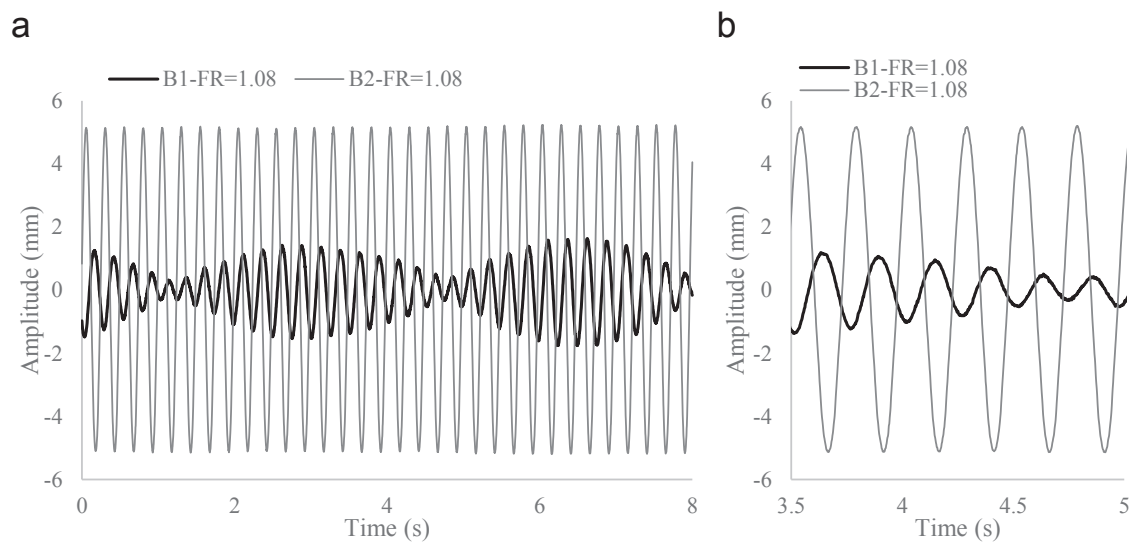


Fig. 13. (a) Time history of the vertical displacements and (b) zoom image at $V/f_2D_2 = 9.4$ and $FR = 1.08$.

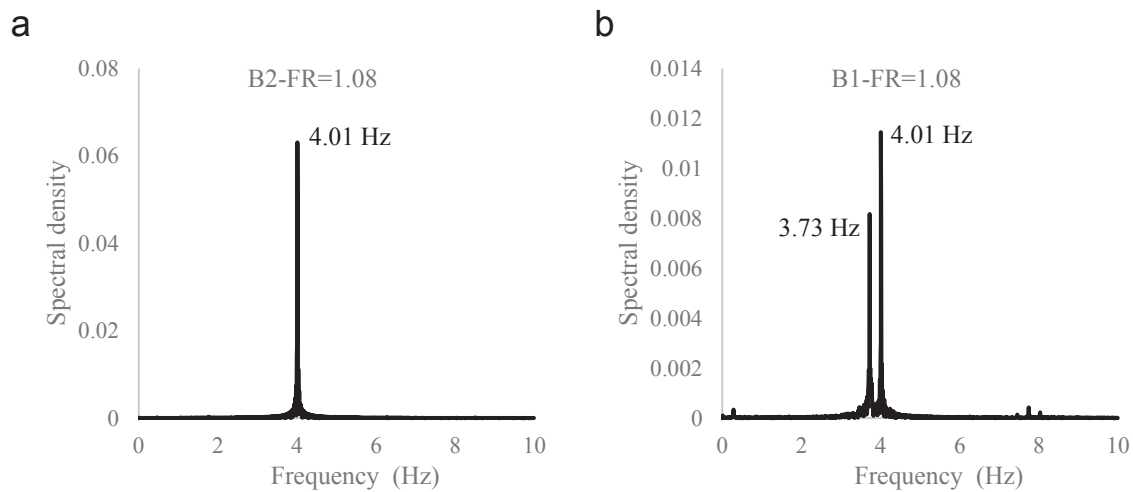


Fig. 14. Spectral densities of the normalized vertical displacements of (a) B2 and (b) B1 at $V/f_2D_2 = 9.4$ and $FR = 1.08$.

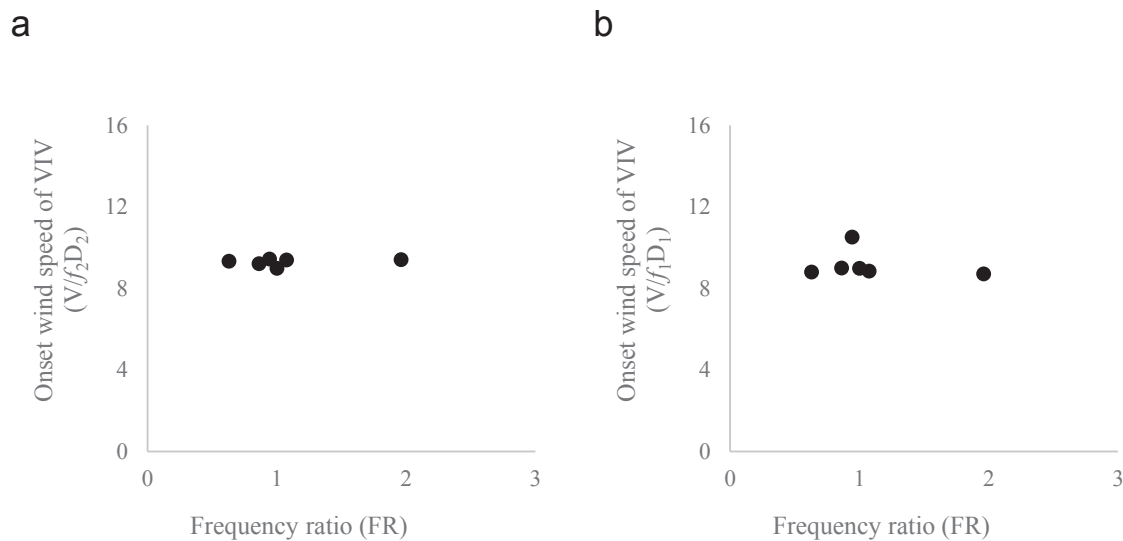


Fig. 15. Onset wind velocities of VIV for a given FR in (a) B2 and (b) B1.

4. Conclusions

The effect of the relative differences in the natural frequencies of parallel bridges was investigated in wind tunnel testing. The differences in the natural frequencies of the two decks are represented in terms of the frequency ratio (FR), which is defined as the ratio of the frequency of the upstream B2 to that of the downstream B1. Six levels of FR were examined to cover all the potential ranges of differences in natural frequencies between the two decks, including two extreme cases of fully separated lock-in wind velocity ranges.

When the ranges of lock-in wind velocity for the two decks were sufficiently separated by a difference in natural frequency, each deck showed independent motions without complex interactions. In fact, the

motions of vortex-induced vibration (VIV) in B2 were almost consistent regardless of FR, while those for B1 showed variations in amplitude as well as in the range of lock-in wind velocity. Even though the upstream deck was not much influenced by the motion of B1, its existence was critical in producing a magnified VIV in B2. On the contrary, the motions of B1 were severely influenced by those of B2. When additional wind tunnel tests were performed with fixed conditions for B2, the motion of B1 also showed a consistent pattern regardless of FR. Accordingly, it can be concluded that a complicated interactive VIV in B1 was induced due to the motion-induced vortices generated from B2. B1 reached its maximum VIV motion when the frequencies of the two decks were identical. This unique condition was initiated during a phase wherein the motion-induced vortices generated from B2 excited B1.

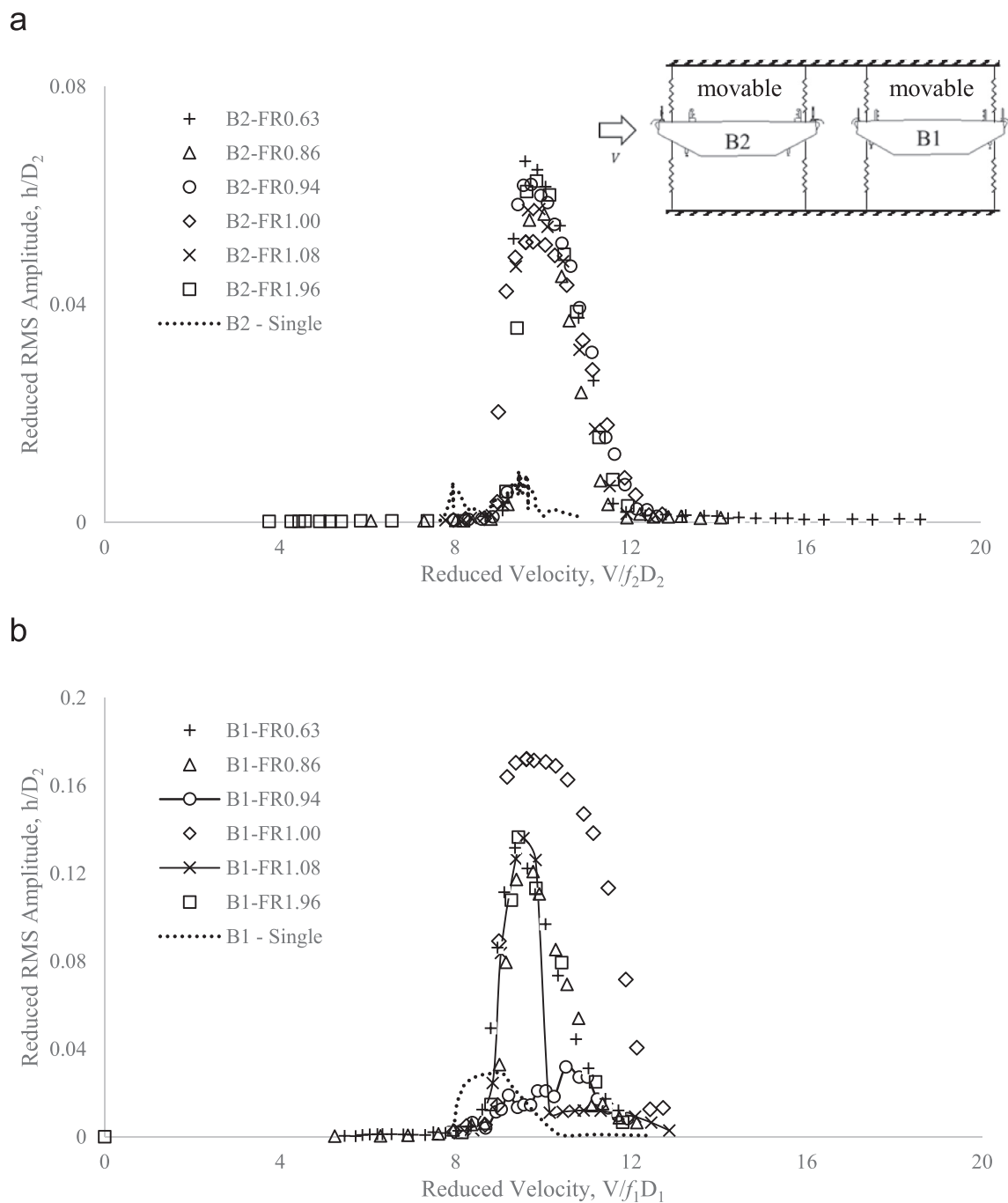


Fig. 16. Amplitude-velocity curves for (a) B2 and (b) B1.

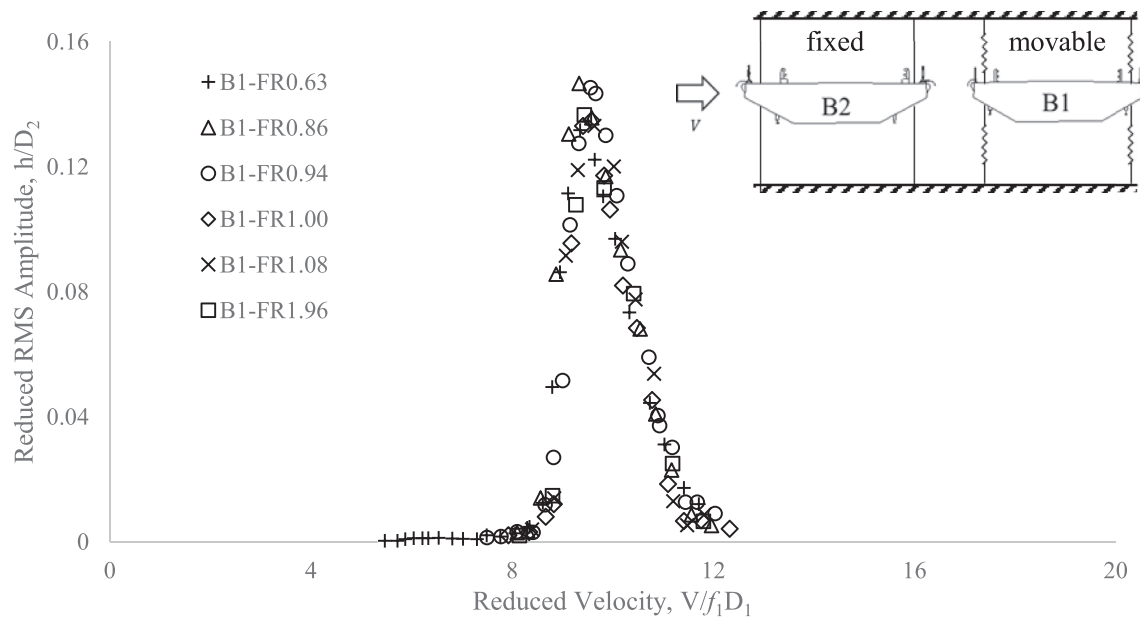


Fig. 17. Amplitude-velocity curves of B1 with fixed B2.

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